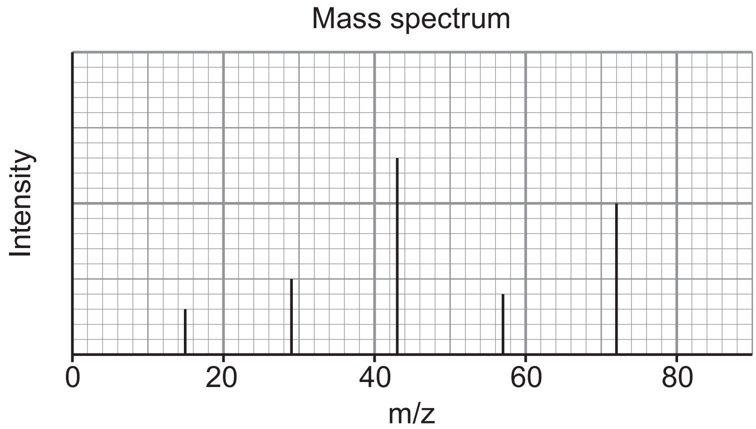
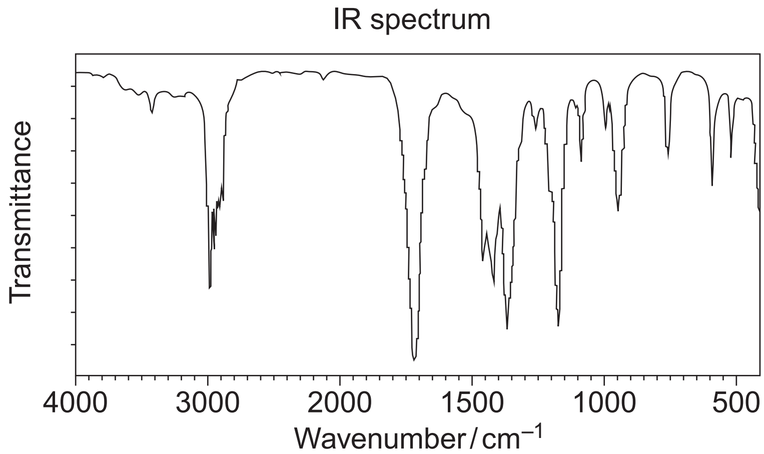
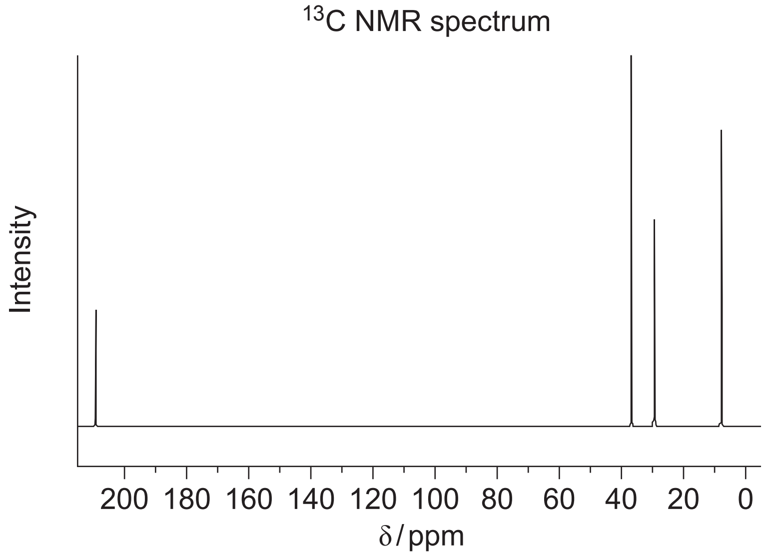
11. (a) Compound **X** contains carbon, hydrogen and oxygen only. It has no reaction with acidified potassium dichromate.

Simplified versions of its mass spectrum, IR spectrum and ^{13}C NMR spectrum are shown.







Examiner only

Identify compound **X**.

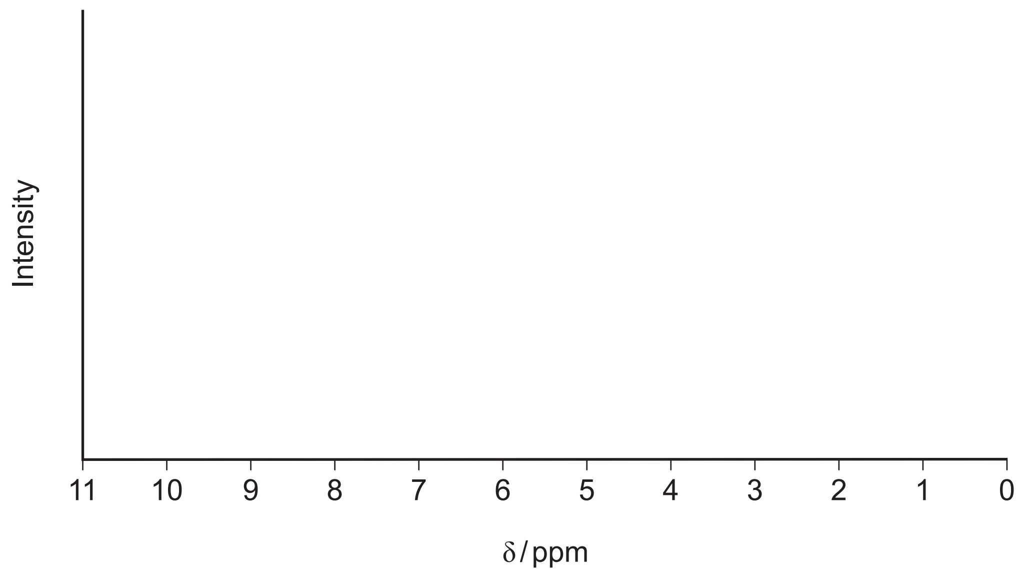
You **must** use information from **all** the sources given and explain how you used it. [8]

Compound **X**

Examiner only

(b) On the axes below, complete the low resolution ^{1}H NMR spectrum you would expect for the compound you identified in part (a).

You should indicate where you would expect to see peaks **and** the relative intensities of the peaks. [2]



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